Optimization & Machine Learning M

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Context and Objective

You have recently joined the Data Operations team at *Capital Bikeshare*, a public bicycle sharing system in Washington D.C. The company is facing a major logistical challenge: customer dissatisfaction due to empty stations (unable to rent a bike) and completely full stations (unable to return a bike). In parallel, Capital Bikeshare is rolling out a fleet of electric bikes (e-bikes) across the network, which requires the strategic placement of charging hubs to keep the bike fleet operational throughout the day.

Your manager has entrusted you with a two-part mission:

- **Machine Learning (Prediction):** Develop a model to predict the hourly demand for bike rentals based on temporal and meteorological factors.
- **Operations Research (Optimization) [Bonus]:** Plan the deployment of charging infrastructure to support the company's growing e-bike fleet.

Deliverables: A clean, commented, and well-structured Jupyter Notebook (`.ipynb`) containing your code and textual analyses.

Machine Learning Tasks

1. Data Exploration and Preprocessing (40%)

In the directory `data`, you are provided with two files: the dataset `bike_sharing_dataset.csv` and a text file `about_bike_sharing_dataset.txt` that describes the dataset's characteristics. Note that the target variable you need to predict is `cnt`.

The raw dataset may contain imperfections and anomalies. You must clean and prepare the dataset for model training.

(a) **Exploration:** Load the dataset. Study its dimensions and understand the meaning of each characteristic.

(b) **Missing Values:** Identify and handle missing values logically. Depending on the nature of the missing periods, they require different imputation strategies.

(c) **Anomalies:** Identify physically impossible values and outliers. Correct, impute, or delete them appropriately. *Hint: Check that characteristics only take physically valid values. Columns can help you correct anomalies in other columns.*

(d) **Feature Selection:** Identify and remove redundant columns, collinearities, and data leakage variables. Justify your choices.

(e) **Feature Engineering:** Encode categorical variables appropriately. *Hint: Apply one-hot encoding to nominal categorical variables. Apply sine/cosine transformations to cyclic variables.*

(f) **Analysis:** Visualize the distribution of the target variable (`cnt`) and its relationship with temporal (hour, day, month) and meteorological variables.

2. Model Development and Evaluation (40%)

(a) **Data Splitting:** Split the dataset into training (80%) and testing (20%) subsets.

(b) **Baseline Model:** Train a baseline linear model (e.g., Linear Regression). Evaluate its performance using suitable metrics (e.g., RMSE, MAE, R-squared).

(c) **Advanced Modeling:** Choose and train an appropriate non-linear machine learning algorithm (e.g., Random Forest).

(d) **Fine-tuning:** Optimize your best model's hyperparameters using Grid Search to improve performance (e.g., explore the number of estimators, the maximum tree depth, and the minimum number of samples required to split a node).

3. Prediction and Interpretation (20%)

(a) **Evaluation:** Use the trained model to make predictions on the testing data and assess its final performance.

(b) **Feature Importance:** Extract and visualize the most important features of your model. Do these align with your initial exploratory data analysis?

(c) **Conclusion:** Summarize your findings. Provide business recommendations.

Operations Research Task

4. E-Bike Charging Hub Placement (Bonus)

Capital Bikeshare is expanding its e-bike fleet across the city. To keep e-bikes operational, the company must install charging hubs at a subset of existing station locations. A charging hub at station i can serve any station j within a coverage radius of D km (i.e., $\text{dist}(i, j) \leq D$). Your task is to determine the **minimum number of stations** to designate as charging hubs so that **every station** in the network is within range of at least one hub.

You have been provided with an `or_data.json` file containing:

- a list of 20 stations;
- a 20×20 symmetrical `distance_matrix` representing travel distances in kilometers between all stations;
- a scalar `coverage_radius` D: the maximum distance (in km) within which a hub can serve a station.

Let $\mathcal{N} = \{1, \ldots, 20\}$ be the set of all stations, and d_{ij} the distance between stations i and j. For each station j, let $\mathcal{C}(j) = \{i \in \mathcal{N} \mid d_{ij} \leq D\}$ be the set of stations that can cover j. Define the binary decision variable:

$$Y_i \in \{0, 1\}, \quad i \in \mathcal{N}: \quad Y_i = 1 \text{ if station } i \text{ is designated as a charging hub.}$$

(a) **Mathematical Formulation:** Write the Binary Integer Program that minimizes the total number of charging hubs opened, subject to the constraint that every station is covered by at least one open hub.

(b) **Implementation:** Import the JSON file and use the `PuLP` library in Python to solve this problem (see the official case studies for guidance). Report the optimal number of hubs p^*, their locations, and which stations each hub covers.

(c) **Sensitivity analysis:** Suppose the operator's budget allows for only $p^* - 1$ hubs. Adapt your model to minimize the number of uncovered stations under this tighter budget, and identify which station(s) lose coverage.